



Original research

The effects of three types of balance training programs on measures of balance and muscle power in prepubertal children: A randomized controlled trial



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ABSTRACT

Objectives: The purpose of this study was to investigate the effects of static (SBT), quasi-dynamic (QDBT), and dynamic (DBT) balance training on balance and muscle power performance in prepubertal children.

Design: Randomized, controlled trial design.

Methods: Fifty-six children (10 and 11 years) were randomly assigned to static ($n = 14$), quasi-dynamic ($n = 14$) and dynamic balance training ($n = 15$), or a control group ($n = 13$). One-legged static (i.e., unipedal stance on stable surface), quasi-dynamic (i.e., unipedal stance on unstable surface), and dynamic balance (i.e., jumping and stabilizing in unipedal stance on unstable surface) were assessed using two dependent variables, i.e., center of pressure surface area (CoP SA) and velocity (CoP V) with opened (OE) and closed (CE) eyes; as well as lower limbs muscle power (i.e., standing long jump test [SLJ] and countermovement jump [CMJ] height).

Results: A significant group $\times$ test interactions were observed for static (CoP SA OE; $p < 0.05$, $d = 1.18$), quasi-dynamic (CoP SA CE; $p < 0.05$, $d = 0.80$), dynamic (CoP V CE; $p < 0.05$, $d = 0.87$) balance and power (SLJ [$p < 0.05$, $d = 0.87$] and CMJ height [$p < 0.01$, $d = 2.66$]) tests. Post hoc analysis showed that DBT demonstrated greater improvement in the measure of balance ($p < 0.05$, $d = 0.80$ to 5.17) and power ($p < 0.05$, $d = 2.66$ to 3.65) performance.

Conclusions: To enhance balance and muscle power performance in prepubertal children, it appears that the level of difficulty in balance training may influence the observed effects.

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Practical implications

- It's worth noting that, because of the relatively dynamic nature of this exercise, DBT exercises may be suggested as a more appropriate tool for younger athletes. Thus, if the goal is to improve balance and muscle power performance in prepubertal children, balance task difficulty appears to moderate the observed effects.
- DBT, consisting of a small number of sets and repetitions (e.g., 2 sets of 30 s on unstable devices), may be recommended as a valuable tool for enhancing balance capabilities and power performance in children.

- Considering that coordination, balance, and power are typically underdeveloped in youth,^{24,26} scientists and practitioners should consider the findings of this study when working with children and implementing balance exercises.

1. Introduction

Evidence-based prescriptions for balance training (BT) in youth have previously been established.^{1–5} It is well known that BT has the potential to induce performance improvements in selected components of physical fitness (e.g., balance, muscle strength/power, change of direction speed).^{3–6} Regarding pediatric population, the relationships between age/puberty and stages of postural balance system development were evidenced.^{7–9} In this context, it was reported that

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children between the ages of four and five years of age employing large and rapid postural adjustment strategies.¹⁰ By the age of seven years, the development of the structures responsible for motor control is complete.¹¹ From 8 to 10 years of age, reductions in speed and increases in precision of adjustments are clearly seen, as there is better integration between the open and closed-loop control systems to maintain this control.^{9,10} According to Roncesvalles et al.,⁸ by the age of 10 years, children will have achieved postural stability similar to adults. Thus, given that balance and coordination are not fully developed in children aged between 10 and 11 years,¹² there is a critical need to foster and enhance balance capabilities during this particular stage.

To maximize the effectiveness of BT, it is crucial to elucidate the optimal dosage of different programming parameters (e.g., training frequency, volume, intensity). While training frequency and volume can easily be manipulated in BT, it is not the case for balance intensity. An often load dimension is “task difficulty”, which may be regarded as an easily implemented tool for quantifying BT intensity.^{3,4,13,14} Indeed, changing the level of balance task difficulty (e.g., varying stance conditions or manipulating the sensory input) has been described as a vehicle to design varied and challenging BT programs. In this context, manipulating the base of support (BoS) and/or sensory inputs (proprioceptive and visual) has been considered as one way for balance task difficulty adaptation.¹⁵ While recent intervention studies have investigated the effects of balance exercises or balance training (BT) with varying task difficulties on static and proactive balance in youth,^{2,5,8} it remains unclear whether diverse static, quasi-dynamic, and dynamic balance training programs yield similar training adaptations in prepubertal individuals.

Additionally, there is evidence of facilitative effects from long-term BT when task difficulty is adequately manipulated.^{4,13,17,18} In this regard, Schedler et al.⁴ reported significant increases in dynamic ($d = 0.35$) and proactive ($d = 0.60$) balance with reduced BoS (i.e., high task difficulty), after 8 weeks of BT using increasing balance task difficulty (high vs low balance task with regard to manipulating the BoS). Similarly, Blasco et al.¹⁶ indicated that varying unstable surfaces, by modifying balance task difficulty, resulted in similar improvement of the static and dynamic domains of balance, and functional performances in young adults. With regard to the aforementioned studies, it seems that there is a lack of an effectiveness mean to manipulate balance training intensity, particularly in prepubertal children.

In addition, a growing number of studies^{19,20} have clearly shown that BT-related adaptations are highly task-specific. However, information about how the type of balance exercise, with increased levels of difficulty (i.e., static, quasi-dynamic, and dynamic type) can moderate the transfer effect to muscle strength and power performance outcomes after the long-term BT is limited. Based on the currently available evidence in the existing literature, we can speculate that increasing balance task difficulty using a variety of dynamic exercise, may yield long-term effects on athletic performance.

Therefore, the purpose of this study was to examine the effects of short-term static (SBT) versus quasi-dynamic (QDBT) versus dynamic (DBT) BT on static, quasi-dynamic, dynamic balance, and lower limbs muscle power in prepubertal children. Based on earlier longitudinal studies^{3,4,16,21} we hypothesized that all BT programs would result in balance and muscle power improvements in prepubertal children. Furthermore, given that prepubertal children rely on a premature postural control system⁵; and that BT-related adaptations are highly task-specific^{4,16,20} we expected that children would benefit more from dynamic BT (high task difficulty), due to the multiple postural control strategies involved in such training.

2. Methods

2.1. Participants

Based on the study of Schedler et al.¹⁸ on the effects of balance training conducted under a high versus a low level of task difficulty on

balance performance, an a priori power analysis with a type I error rate of 0.05 and 80 % statistical power was computed. The analysis indicated that overall, 48 participants would be sufficient to observe significant, large-sized acute effects (Cohen's $f \geq 0.5$) for static balance performance (i.e., static one-legged stance time). In addition, to account for potential loss of participants to attrition, we added 10 % to our total target sample, equating to 53.

Inclusion criteria included healthy, normoweight prepubertal children with a history of regular attendance to physical education lessons (twice a week). Additionally, participants should not be engaged in any after-school activities or formalized strength and conditioning training programs. Finally, 56 prepubertal children, aged between 10 and 11 years, from the same public school, volunteered to participate in this study.

All participants were randomly assigned to three different exercise groups by a staff member not involved in the measurements through a computer-generated method (<https://random.org/lists>). A parallel design with a stratified randomization method according to gender was used. The three groups were: Static Balance Training (SBT; $n = 14$; age = 10.75 ± 0.23 yrs, height = 147.50 ± 5.91 cm, body mass = 41.39 ± 2.39 kg, PHV = -2.26 ± 0.29), Quasi-Dynamic Balance Training (QDBT; $n = 15$; age = 10.70 ± 0.32 yrs, height = 145.87 ± 8.18 cm, body mass = 41.57 ± 2.08 kg, PHV: -2.36 ± 0.41), Dynamic Balance Training (DBT; $n = 14$; age = 10.67 ± 1.8 yrs, height = 148.36 ± 7.71 cm, body mass = 41.29 ± 1.53 kg, PHV = -2.28 ± 0.35), and a control group (CON; $n = 13$; age = 10.65 ± 0.28 yrs, height = 149.54 ± 5.90 cm, body mass = 41.30 ± 1.54 kg, PHV = -2.25 ± 0.30), in order to avoid learning effects and fatigue.

All participating children came from families of similar socioeconomic backgrounds and had identical daily school schedules. All training interventions (i.e., SBT, QDBT, and DBT) were integrated into the two weekly physical education classes of participants. Before the start of the study, participants were given a letter that included written information about the experimental procedure with a parental consent request. Parental and subject informed consent was obtained after a thorough explanation of the objectives and scope, procedures, risks (i.e., pain, injury, illness or disease brought about by the methods and procedures of the current study), and benefits of the study. Participants were free to withdraw at any time if they felt discomfortability or pain during the procedures or they did not want to participate anymore. The study was conducted according to the latest version of the Declaration of Helsinki and the protocol was fully approved by the Local Clinical Research Ethics Committee “Personal Protection Committee” under the following code (N°: 0365//2021) before the commencement of any assessments. None of the participating children had a history of musculoskeletal, neurological, or orthopedic disorders that might have impaired their ability to execute the prescribed tasks/tests. All the procedures were conducted following the Consolidated Standards of Reporting Trials (CONSORT 2022, [Appendix 1](#)) and Better reporting of interventions: template for intervention description and replication (TIDieR, [Appendix 2](#)) checklists.

2.2. Procedures

According to the study aim, all the procedures were conducted at Gafsa (Tunisia) primary school coinciding with the second half of the academic year (April–June 2021). All training interventions (SBT, QDBT, and DBT) were integrated into the two weekly physical education lessons of the participants, which were carried out in the morning, at the same time and with the same staff. The measurements were specific to purpose of trial, they were not part of the typical curriculum contents. The same certified strength and conditioning coaches supervised face-to-face all the testing/intervention procedures sessions, which were conducted in groups. The staff in charge of the measurements and analyses was blinded from the group allocation. The attendance rate was controlled by directly counting the assistants. Before the start of the study, all participants in both the interventions and control groups completed a 2-week familiarization period (3 sessions a week)

to become acquainted with the environment, session structure, equipment, techniques used in each balance and power test, as well as the BT exercises.

Balance (primary outcome) and power (secondary outcome) performance were tested before and following the 6-week SBT (i.e., tandem stance with OE and CE), QDBT (tandem stance on unstable surface with OE and CE) and DBT (i.e., Jumping and standing on tandem stance on balance pad with OE and CE) on period in an indoor sport hall of the local public primary schools. The balance was the primary outcome of the study as it was directly associated with the training interventions. Additionally, jumping performance (power) was selected as a secondary outcome to assess potential positive transfer of balance training to jumping performance. The data collection sessions started 48 h after the familiarization period, where all prepubertal children received an orientation where the static (center of pressure surface with opened eyes [CoP SA OE]), quasi-dynamic (center of pressure surface with closed eyes [CoP SA CE]) and dynamic (center of pressure velocity with closed eyes [CoP V CE]) balance and power (standing long jump [SLJ] and counter-movement jump [CMJ]) tests were explained. Body height, body mass, body composition, balance (static, quasi-dynamic and dynamic balance tests) and muscle power performance of lower limbs (horizontal and vertical jump tests) were assessed. All subjects conducted a standardized 10-min warm-up prior to testing, that consisted of submaximal running (e.g., skipping, hip in/out) and dynamic stretching.

The body height and the body mass were collected using a wall-mounted stadiometer (i.e., Florham Park, NJ) and an electronic scale (i.e., Baly International, West Sussex, England), respectively. The sum of skinfolds was assessed using Harpenden skinfold calipers. Body fat percentage was therefore estimated using the equation of Deurenberg et al.²² (Body fat percentage = $222.23 + 26.56 \log$ (Biceps, triceps, subscapular, and suprailiac)), who reported similar prediction errors between adults and adolescents. Biological maturity was evaluated, non-invasively, by incorporating measures of chronological age and body height into a regression equation to subsequently predict biological age from pic height velocity (PHV), as described by Moore et al.²³

2.2.1. Primary outcome: balance performance

Balance tests were performed with eyes opened (EO) and closed (EC) using a force plate with three strain gauges and a sampling rate of 40 Hz (PostureWin®, Techno Concept®, Cereste, France) according to the procedure described by Hammami et al.¹⁷ Throughout all balance tests, participants were instructed to look straight ahead

at a cross, placed at eye level on a nearby wall (2 m distance). Each test trial lasted 30 s. As dependent variables, two center of pressure (CoP) sway parameters were analyzed: COP surface area (CoP SA) in mm² and CoP velocity (CoP V) in mm/ms). Similarly, each test trial lasted 30 s. The domains namely static (center of pressure surface with opened eyes [CoP SA OE]), quasi-dynamic (center of pressure surface with closed eyes [CoP SA CE]) and dynamic (center of pressure velocity with closed eyes [CoP V CE]) balance were analyzed. CoP SA represents the ellipse of the area covered by the trajectory of the CoP whereas the CoP V indicates the total distances covered by the CoP divided by the duration of the sampled period.⁴ For the balance parameters, the lower the value, the better the balance performance.²⁴ Two trials were performed with approximately 2 min of recovery and the best result was used for analysis.

2.2.1.1. Static balance test. For the static balance test, participants stood in a single leg stance on the dominant leg for 30 s with EO, on a stable surface. The leg used to kick a soccer ball identified the dominant leg²⁵ (Fig. 1, a).

2.2.1.2. Quasi-dynamic balance test. For the quasi-dynamic balance test, participant stood on the stabilometric platform with unipedal leg stance on the dominant leg with EC, on unstable surface (i.e., BOSU) (Fig. 1, b).

2.2.1.3. Dynamic balance test. For the dynamic balance test, participants jumped with EC onto the stabilometric platform where a BOSU ball was placed upside-down and stabilized the dominant leg on the unstable surface (i.e., BOSU) (Fig. 1, c).

2.2.2. Secondary outcome: muscle power performance

Each participant performed a bilateral standing horizontal long jump (SLJ) test where the participants stood stationary with the toes aligned level with the start line and were instructed to push off vigorously and jump forward as far as possible. Participants were allowed the use of a counter-movement with arms and body swing. The distance (cm) jumped from the start line at takeoff to the position of the heel upon landing was measured in centimeters using a metal tape measure. Participants completed three trials, with the highest score being recorded.

The counter-movement jump (CMJ) test was performed using an Ergojump® system (ErgojumpP apparatus, Globus Italia, Codogne, Italy), which recorded jump height (cm). Each test was separated by a

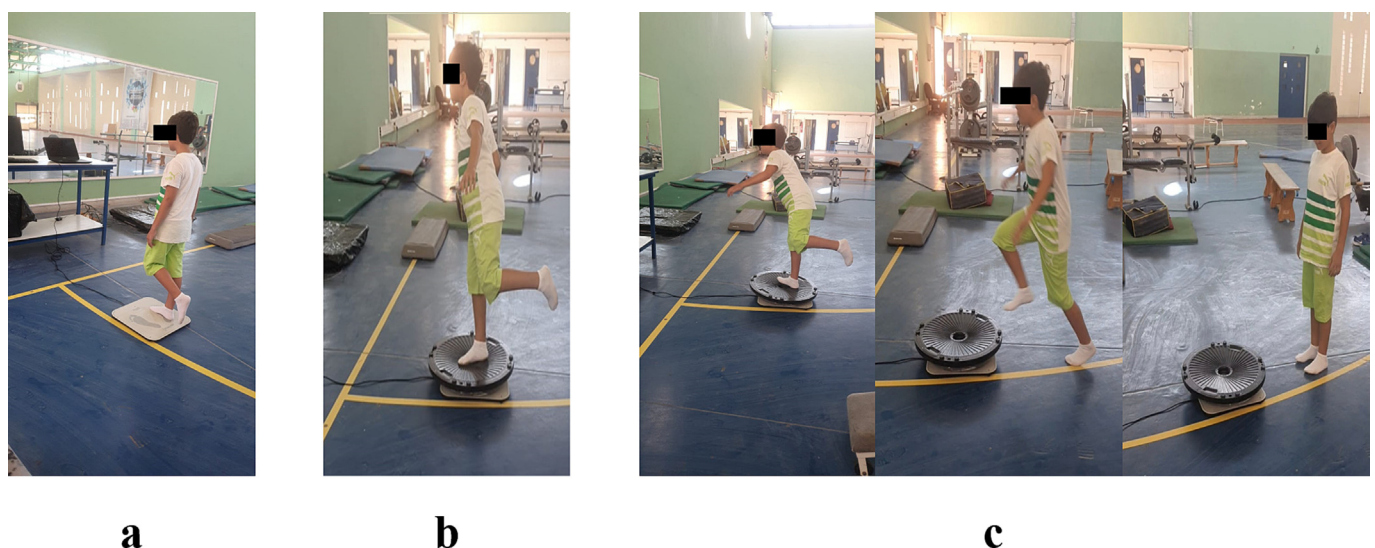


Fig. 1. Schematic description of the three balance exercise types. a Static balance test: unipedal stance on stable surface, b quasi-dynamic balance test: unipedal stance test on unstable surface, and c dynamic balance test: jumping and stabilizing in unipedal stance on unstable surface.

passive recovery period of at least 5 min. The CMJ involved the participants descending from an upright standing position until a knee flexion angle of 90°, followed immediately by a vertical jump. Participants were encouraged to perform the eccentric phase of the jump as quickly as possible to maximize jump height. Two trials were performed, with approximately 2 min recovery, and the best result was used for analysis.

2.2.3. Training programs

After, assessment, all experimental groups conducted 6 weeks of BT with 2 sessions per week (30–35 min each) that included six balance exercises per training session (Table 1). The duration and frequency of

the programs were based on a previous study that found significant improvements in balance with large to very large effect sizes of 6 weeks and frequency between 1 and 3 days per week.³ After a 15-min warmup program, including general (e.g., skipping, running with hip in/out) and specific (e.g., two-/one-legged stance on unstable devices, forward/backward beam walking) exercises, two sets per balance exercise were performed for 30 s each with a 60 s rest period between sets and a 90 s break between exercises. While participants of the SBT group performed all balance exercises with a low level of task difficulty, subjects in the QDBT and DBT groups executed the same or similar exercises using a quasi-dynamic and dynamic high level of task difficulty,

Table 1
Description of the static, quasi-dynamic and dynamic balance-training programs.

Week	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Sets × repetition	1 × 8	2 × 10	3 × 10	1 × 10	3 × 10	2 × 10
SBT	a) Tandem stance on stable surface with eyes opened and on right leg. b) Tandem stance on stable surface with eyes opened and on left leg. c) Tandem stance with doing quick cutting motions of the hands in sagittal plane on stable surface on right leg and with eyes opened. d) Tandem stance with doing quick cutting motions of the hands in sagittal plane on stable surface on left leg and with eyes opened.	a) Tandem stance on stable surface with eyes closed and on right leg. b) Tandem stance on stable surface with eyes closed and on left leg.	a) Tandem stance with doing quick cutting motions of the hands in sagittal plane on stable surface and right leg and with eyes closed. b) Tandem stance with doing quick cutting motions of the hands in sagittal plane on stable surface and left leg and with eyes closed.	a) Single leg stance on right leg and eyes opened with doing quick cutting motions of the hands in sagittal plane on stable surface. b) Single leg stance on left leg and eyes opened with doing quick cutting motions of the hands in sagittal plane on stable surface.	a) Standing on left leg stance on stable surface with eyes closed.	a) Standing on right leg stance on stable surface with eyes closed and doing cutting motions of the hand in sagittal plane. b) Standing on left leg stance on stable surface with eyes closed and doing cutting motions of the hand in sagittal plane.
QDBT	a) Tandem stance on balance pad with eyes opened and on right leg. b) Tandem stance on balance pad with eyes opened and on left leg. c) Tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad on right leg and with eyes opened. d) Tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad on left leg and with eyes opened.	a) Tandem stance on balance pad with eyes closed and on right leg. b) Tandem stance on balance pad with eyes closed and on left leg.	a) Tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad and right leg and with eyes closed. b) Tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad and left leg and with eyes closed.	a) Single leg stance on right leg and eyes opened with doing quick cutting motions of the hands in sagittal plane on balance pad . b) Single leg stance on left leg and eyes opened with doing quick cutting motions of the hands in sagittal plane on balance pad .	a) Standing on right leg stance on balance pad with eyes closed. b) Standing on left leg stance on balance pad with eyes closed.	a) Standing on right leg stance on balance pad with eyes closed and doing cutting motions of the hand in sagittal plane. b) Standing on left leg stance on balance pad with eyes closed and doing cutting motions of the hand in sagittal plane.
DBT	a) Jumping and standing on tandem stance on balance pad with eyes opened and on right leg. b) Jumping and standing on tandem stance on balance pad with eyes opened and on left leg. c) Jumping and standing on tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad on right leg and with eyes opened. d) Jumping and standing on tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad on left leg and with eyes opened.	a) Jumping and standing in tandem stance on balance pad with eyes closed and on right leg. b) Jumping and standing in tandem stance on balance pad with eyes closed and on left leg.	a) Jumping and standing in tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad and right leg and with eyes closed. b) Jumping and standing on tandem stance with doing quick cutting motions of the hands in sagittal plane on balance pad and left leg and with eyes closed.	a) Jumping and standing with single leg stance on right leg and eyes opened with doing quick cutting motions of the hands in sagittal plane on balance pad . b) Jumping and standing with single leg stance on left leg and eyes opened with doing quick cutting motions of the hands in sagittal plane on balance pad .	a) Jumping and standing on right leg stance on balance pad with eyes closed. b) Jumping and standing on left leg stance on balance pad with eyes closed.	a) Jumping and standing on right leg stance on balance pad with eyes closed and doing cutting motions of the hand in sagittal plane. b) Jumping and standing on left leg stance on balance pad with eyes closed and doing cutting motions of the hand in sagittal plane.

Note: Bold font indicates differences in task difficulty. SBT, balance training using a static balance exercise task; QDBT, balance training using a quasi-dynamic balance exercise task; DBT, balance training using a dynamic balance exercise task.

respectively (Table 1). However, all groups executed the same training volume (i.e., number of exercises, number of sets per exercise). Training progression was applied in all exercise groups, for instance changing from bipedal stance to step, tandem, and unipedal stance. In addition, the visual input was manipulated as the children progressed over the training period (e.g., eyes opened vs. closed). SBT, QDBT, and DBT replaced portions of the regular physical education curriculum. The standard physical education class lasted for 60 min and included a 15-minute warmup followed by various activities such as running, jogging, playing ball, and engaging in competitive games promoting physical movement, such as basketball, handball, and football. During the intervention, the control group participated in traditional outdoor physical education games throughout the intervention period, which included free games with fewer defined rules and techniques, such as catch and cook and hide-and-seek. The training volume was consistent across all four groups. Each training session concluded with a 15-minute cool-down, incorporating flexibility exercises and low-intensity jogging. All training sessions ended with a 15-min cool-down, that included flexibility exercises and jogging at a low intensity.

3. Data analysis

Data are presented as means and standard deviations (SD). After a basic data curation, normality was assessed and confirmed using the Shapiro–Wilk test. No outcome data were missing or excluded from the study. The data were then analyzed using a 4 (groups: SBT, NDBT, DBT, CON) by 2 (time: pre, post) analysis of variance (ANOVA) for repeated measures. Where the assumption of sphericity was violated, Greenhouse–Geisser correction was used to interpret the results. Additionally, effect sizes (ES) were determined from ANOVA output by converting partial eta-squared to Cohen's *d*. Moreover, within-group ES were computed using the following equation: $ES = (\text{mean post} - \text{mean pre}) / SD$. In accordance with Hopkins,²⁶ ES were considered to

be either 'trivial' (<0.2) 'small' (>0.2–0.6), 'moderate' (>0.6–1.2), 'large' (>1.2–2), or 'very large' (>2). Test re-test reliability of the variables was assessed using Cronbach's model of ICCs and SEMs according to the method of Peltola.²⁷ The level of significance was set, a priori, at $p < 0.05$. The statistical analysis was carried out using IBM SPSS (version 25, IBM Corp., Armonk, NY, United States) and R Core Team³³.

4. Results

The sample was composed by 56 prepubertal children, including 31 males (68.42 %) and 25 females (31.58 %). As planned, all of them completed the study (Fig. 2). The rate of attendance among the participants was 100 %. No missing sessions were recorded. No training or testing-related adverse events were reported. There were no statistically significant differences in baseline values between groups. In addition, no significant between-group pre-training differences were found for any of the analyzed parameters. Participants' characteristics are included in Table 2.

4.1. Reliability of the applied tests

Intra-class correlation coefficients (ICCs) showed good reliability for all tests. ICC-values ranged from 0.79 to 0.95, with a standard error of measurement (SEM) from 0.47 to 12.03 and a coefficient of variation of <5 %. Furthermore, a paired *t*-test showed no significant differences between the scores recorded during the two trials for all measured variables.

4.2. Primary outcome: effects on balance performance

4.2.1. Static balance performance

A significant main effect of time was found for the static balance test parameter (CoP SA OE) ($p < 0.001$, $d = 1.80$). In addition, a significant group $\times$ time interactions were detected for the CoP SA OE ($p < 0.05$,

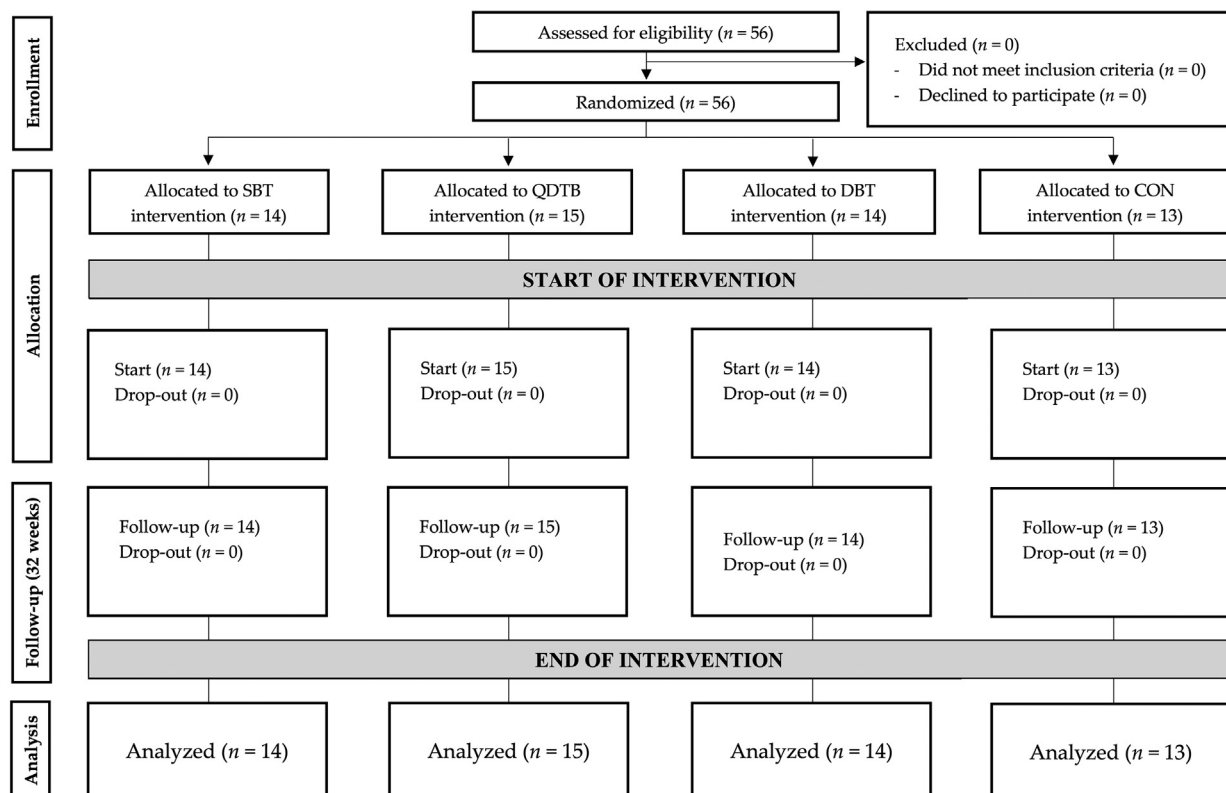


Fig. 2. Participant flow through the study.

Table 2
Anthropometric characteristics of the study participants.

	SBT (n = 14)	QDBT (n = 15)	DBT (n = 14)	CON (13)	ANOVA P-value
Age (yrs)	10.75 ± 0.23	10.70 ± 0.32	10.67 ± 1.8	10.65 ± 0.28	0.77
Height (cm)	147.50 ± 5.91	145.87 ± 8.18	148.36 ± 7.71	149.54 ± 5.90	0.57
BM (kg)	41.39 ± 2.39	41.57 ± 2.08	41.29 ± 1.53	41.30 ± 1.54	0.98
BF%	6.73 ± 0.35	6.74 ± 0.32	6.86 ± 0.24	6.90 ± 0.15	0.27
PHV	−2.26 ± 0.29	−2.36 ± 0.41	−2.28 ± 0.35	−2.25 ± 0.30	0.82
APHV	13.03 ± 0.22	13.06 ± 0.30	12.95 ± 0.27	12.90 ± 0.24	0.33

1) Values are means and standard deviations (SD); SBT, Static Balance Training group; QDBT, Quasi Dynamic Balance Training group; DBT, Dynamic Balance Training group; CON, control group; BM, body mass; BF%, body fat percent; PHV, peak height velocity; APHV, years at peak height velocity.

$d = 1.18$) (Fig. 3). Post hoc analyses showed very large improvements in CoP SA OE performance from pretest to posttest in favor of the DBT ($p < 0.01$, $d = 5.17$) with contrast to the QDBT ($p < 0.01$, $d = 2.55$) and the SBT ($p < 0.01$, $d = 2.21$). However, the CG did not reveal any significant improvement from pretest to posttest ($p = 0.13$, $d = 0.77$) (Table 4).

4.2.2. Quasi-dynamic balance performance

A significant main effect of time was found for the quasi-dynamic balance variable (CoP SA CE) ($p < 0.001$, $d = 1.21$). A significant group $\times$ time interactions were detected for the CoP SA CE only ($p < 0.04$, $d = 0.80$). Post hoc analyses indicated very large improvement ($p = 0.001$, $d = 2.55$) in CoP SA CE performance from pretest to posttest in favor of the DBT. However, the CG did not reveal any significant improvement from pretest to posttest ($p = 0.43$, $d = 0.46$) (Table 4).

4.2.3. Dynamic balance performance

A significant main effect of time was established for the dynamic balance variable (CoP V CE) ($p < 0.001$; $d = 1.56$). A significant group $\times$ time interactions were found for CoP V CE ($p < 0.05$, $d = 0.87$). Post hoc analysis showed very large improvements in CoP V CE performance from pretest to posttest in favor of the DBT group ($p < 0.01$, $d = 2.35$) with comparison to the QDBT ($p < 0.01$, $d = 2.25$) and the SBT ($p < 0.01$, $d = 2.16$) groups. However, the CG did not reveal any significant improvement from pretest to posttest ($p = 0.82$, $d = 0.40$) (Table 4).

4.3. Secondary outcome: effects on muscle power performance

4.3.1. Standing long jump performance

Statistical calculations revealed a significant main effect of time for SLJ ($p < 0.01$, $d = 2.84$) and a significant group $\times$ test interaction ($p < 0.01$, $d = 1.70$) (Fig. 4). Post hoc analysis showed very large

improvement in SLJ performance from pretest to posttest in favor of the DBT group ($p < 0.01$, $d = 3.55$). However, the CG did not reveal any significant improvement from pretest to posttest ($p > 0.05$, $d = 0.02$) (Table 4).

4.3.2. Countermovement jump performance

A significant main effect of time was established for the CMJ height ($p = 0.0001$; $d = 4.76$) and a significant group $\times$ time interactions ($p = 0.0001$, $d = 2.66$) were also noted. Post hoc analysis showed very large enhancements from pretest to posttest in favor of the DBT group ($p < 0.01$, $d = 6.78$). However, the CG did not reveal any significant improvement from pretest to posttest ($p > 0.05$, $d = 0.45$) (Table 4).

5. Discussion

This study was an investigation of the effects of three 6-weeks BT programs, involving a static (SBT), quasi-dynamic (QDBT), and dynamic (DBT) BT exercises on static (CoP SAE, primary outcome), quasi-dynamic (CoP SA CE, secondary outcome), and dynamic (CoP V CE, secondary outcome) balance and power (SLJ and CMJ, secondary outcome) performance in children. Based on similar cross-sectional^{3,14,28,29} and longitudinal^{4,16,21} studies and with regard to our research hypothesis, the main finding was that DBT would show the largest benefits as the DBT tasks are highly dynamic compared with QDBT, SBT, and CON on balance (primary outcome) and power (secondary outcome) parameters. Therefore, it is likely that similar to young adult²⁹ and adolescent,^{4,30} children responded positively to a bout of DBT exercise, which consequently improved their balance and power capacities more than when performing slow balance exercises intensity (i.e., QDBT and SBT). This study was the first to report the longitudinal effects of increasing balance task difficulty on the measure of static, quasi-dynamic, and static balance and power performance in children being aged 10–11 years old.

Partly in accordance with our hypothesis, we detected greater improvements in the main outcome variable (i.e., balance performance, CoP SA and CoP V) for the group that used a high (i.e., DBT) compared to quasi (QDBT) and low (i.e., SBT) balance task difficulty during BT. There is empirical evidence^{4,21,29,30} that balance performance can be challenged by the level of balance task difficulty. In this regard, Schedler et al.⁴ showed that following 7 weeks of BT, BT-high appears to be more effective for increasing measures of proactive (Y-Balance Test [YBT] reach distance, Functional Reach Test [FRT]; Timed-Up-and-Go Test [TUG]) balance ($\Delta + 8\%$, $p < 0.001$, $d = 0.35$) in adolescent. Similar, Murillo et al.²⁹ assessed the effect of increasing difficulty in a standing balance task in young adults. The amplitude of postural sway significantly increased as the level of task difficulty changed from standing on a static surface over standing on a surface with medium followed by high instability. Further, Blasco et al.¹⁶ determined the effect of BT with different stability conditions on balance performance in young adults. Participants were assigned to BT groups that trained on stable ground (corresponds to BT-low) or on unstable surfaces (corresponds

Table 3
Test-retest reliability of the applied power and balance tests.

	ICC _{3,1} (95 % CI)	SEM	CV (%)
Static balance test			
CoP SA OE (mm ²)	0.91 (0.84–0.94)	12.03	0.77
Quasi-dynamic balance test			
CoP SA CE (mm ²)	0.95 (0.91–0.97)	5.73	0.28
Dynamic balance test			
CoP V CE (mm/ms)	0.79 (0.65–0.88)	5.16	2.68
Power tests			
SLJ (cm)	0.93 (0.87–0.96)	1.31	0.85
CMJ height (cm)	0.81 (0.68–0.89)	0.47	2.07

ICC, intra-class correlation coefficient; SEM, standard error of measurements; CV (%), coefficient of variation percent; CoP SA OE, center of pressure surface area with opened eyes; CoP SA CE, center of pressure surface area with closed eyes; CoP V CE, center of pressure velocity with closed eyes; SLJ, standing long jump; 3 hop jump; CMJ height, countermovement jump height.

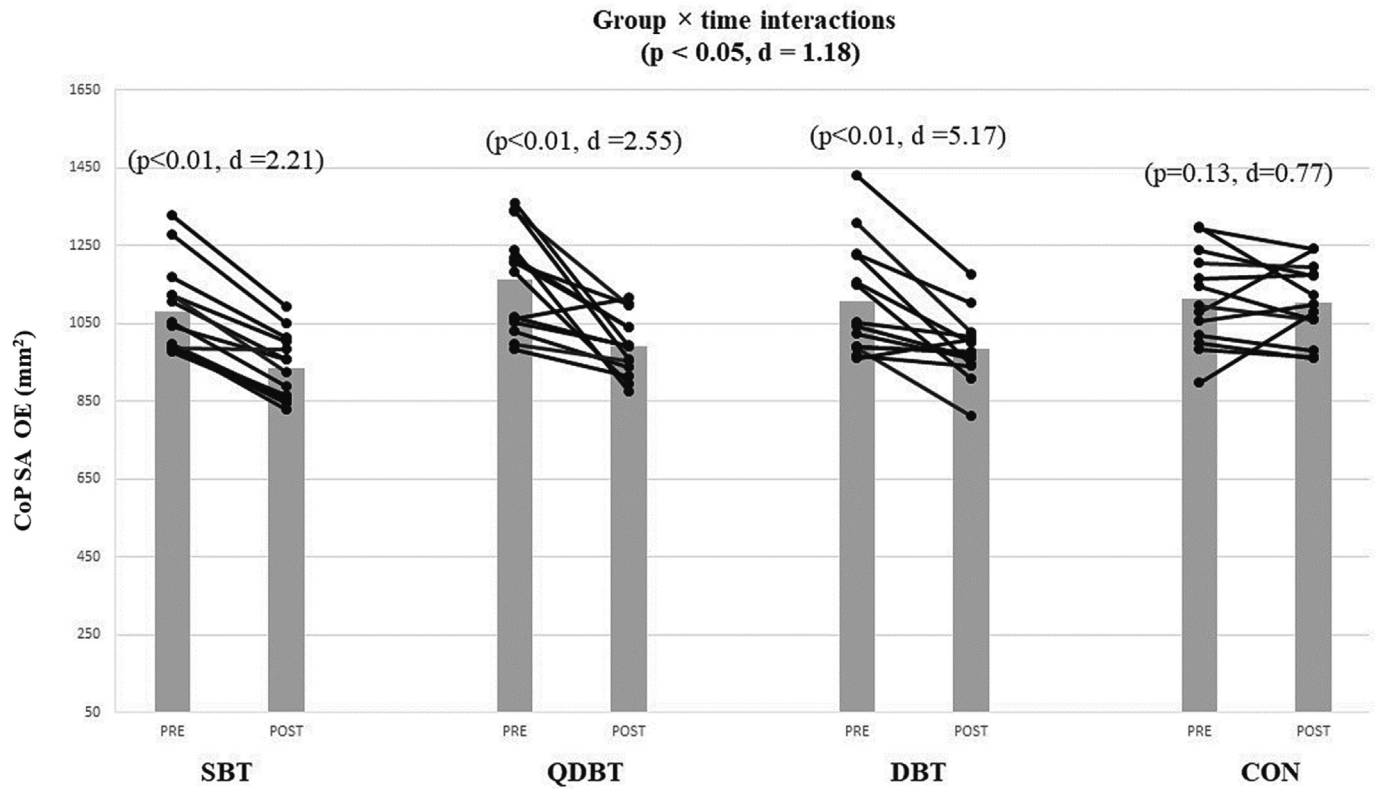


Fig. 3. Individual and mean pre- and post-testing data of static balance performance (CoP SA OE) by SBT, QDBT DBT and CON. SBT = static Balance Training group; QDBT = Quasi-Dynamic Balance Training group; DBT = Dynamic Balance Training group; CON = control group; d: effect size.

to BThigh). Following 3 weeks of treatment (3 sessions per week), both groups improved their balance performance (i.e., YBT and FRT reach distance, one-legged stance time).

Thus, regularly performing balance tasks of dynamic as compared to quasi-dynamic and static BT allows larger adaptive processes, which explains the larger improvements we observed in the DBT compared to

Table 4

Effects of static, quasi-dynamic and dynamic balance training programs on measures of balance and power performance in children.

Variables	Groups	Pre-intervention	Post-intervention	Δ %	Cohen's <i>d</i> (Pre- vs -post)	ANOVA P-value (Cohen's <i>d</i>)		
						Time	Group	Group × time
Balance performance								
Static balance test								
CoP SA OE (mm ²)	SBT	1082.12 ± 112.55	936.92 ± 83.66	−13.21 ± 4.6	−1.29	0.0001 (1.80)	0.104 (0.35)	0.001 (1.18)
	QDBT	1153.64 ± 131.17	990.29 ± 74.83	−13.80 ± 10.4				
	DBT	1111.78 ± 143.80	1006.42 ± 108.01	−10.15 ± 9.0				
	CON	1116.54 ± 128.87	1091.94 ± 93.44	−0.38 ± 9.0				
Quasi-dynamic balance test								
CoP SA CE (mm ²)	SBT	2241.92 ± 198.09	2153.32 ± 214.78	−3.45 ± 11.2	−0.45	0.0001 (1.21)	0.006 (0.52)	0.04 (0.80)
	QDBT	2271.01 ± 198.54	2040.15 ± 222.67	−10.41 ± 7.3				
	DBT	2191.84 ± 139.62	2111.75 ± 215.07	−4.01 ± 7.6				
	CON	2398.48 ± 154.10	2359.57 ± 241.69	−1.55 ± 6.6				
Dynamic balance test								
CoP V CE (mm/mms)	SBT	89.07 ± 14.89	88.52 ± 17.02	−0.36 ± 15.4	−0.93	0.0001 (1.56)	0.86 (0.11)	0.02 (0.87)
	QDBT	93.87 ± 10.96	83.74 ± 6.71	−10.09 ± 9.1				
	DBT	89.82 ± 9.18	82.10 ± 9.19	−8.64 ± 7.4				
	CON	94.59 ± 11.39	82.58 ± 13.60	−12.70 ± 12.3				
Power performance								
SLJ (cm)	SBT	159.21 ± 15.19	178.36 ± 23.58	11.92 ± 8.8	1.26	0.0001 (2.84)	0.12 (0.34)	0.0001 (1.70)
	QDBT	161.93 ± 15.55	184.33 ± 15.39	14.86 ± 7.8				
	DBT	160.79 ± 16.61	187.86 ± 20.73	17.99 ± 11.0				
	CON	159.67 ± 16.06	160.17 ± 16.10	0.01 ± 3.2				
CMJ height (cm)	SBT	12.65 ± 1.00	14.99 ± 1.02	18.77 ± 7.2	2.33	0.0001 (4.76)	0.001 (0.64)	0.001 (2.66)
	QDBT	12.34 ± 1.23	14.59 ± 1.06	19.03 ± 6.8				
	DBT	12.79 ± 0.97	14.53 ± 0.98	14.18 ± 4.9				
	CON	12.24 ± 1.03	12.23 ± 0.83	0.77 ± 6.6				

Note: SBT, Static Balance Training group; QDBT, Quasi-Dynamic Balance Training group; DBT, Dynamic Balance Training group; CON, Control group; CoP SA OE, center of pressure surface area with opened eyes; CoP SA CE, center of pressure surface area with closed eyes; CoP V CE, center of pressure velocity with closed eyes; SLJ, standing long jump; CMJ height, counter-movement jump height.

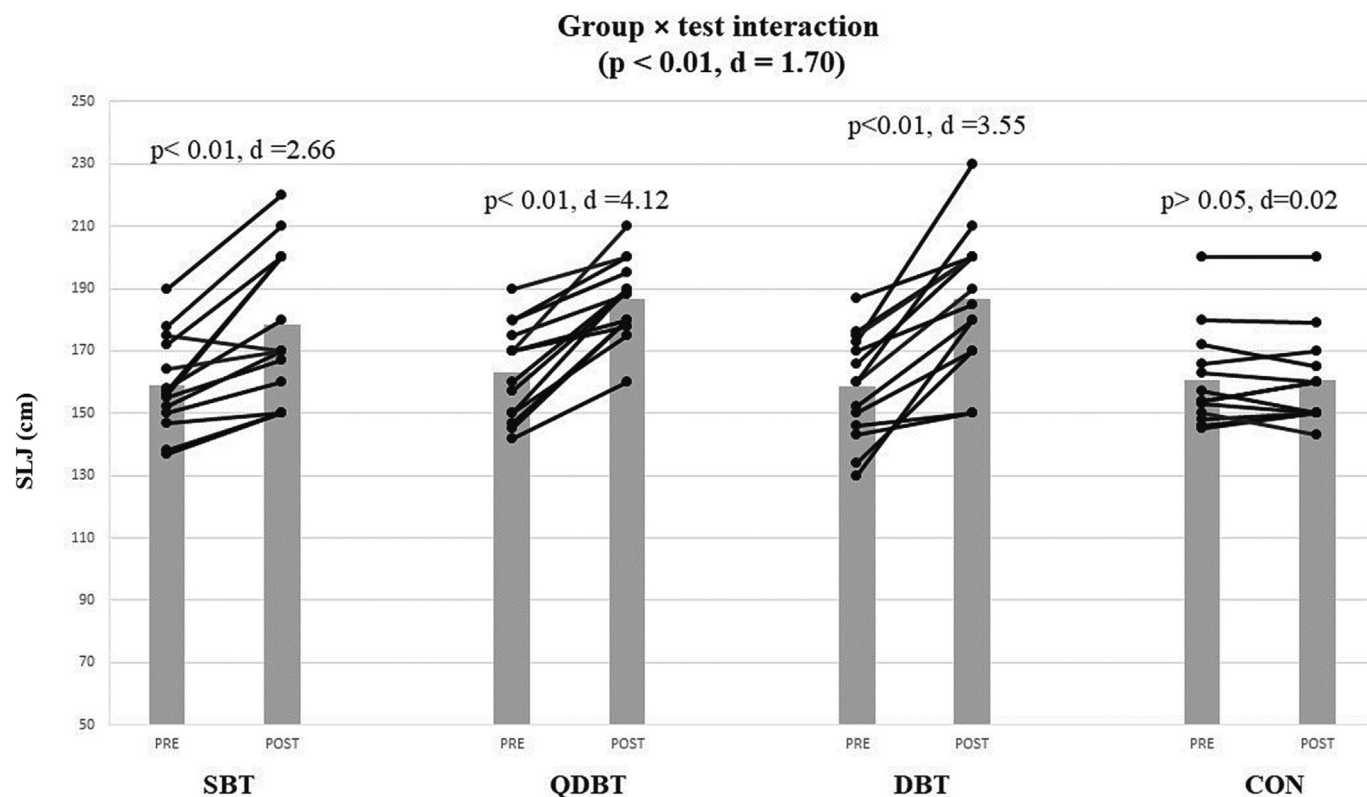


Fig. 4. Individual and mean pre- and post-testing data of power performance (SLJ) by SBT, QDBT DBT and CON. SLJ: standing long jump, SBT = static Balance Training group; QDBT = Quasi-Dynamic Balance Training group; DBT = dynamic Balance Training group; CON = control group; d: effect size.

the QDBT and SBT groups. Furthermore, DBT may elicit larger changes in sensory integration and postural control strategy compared to QDBT and SBT. In this regard, Gebel et al.³⁰ showed that the activity of selected brain areas (e.g., bilateral central) was affected by the increasing of the postural demands. Additionally, following 6 weeks of balance training on an unstable surface (i.e., slackline), Giboin et al.²¹ reported that the decrease of the H-reflex or the Hoffmann's reflex (which is defined by the reflexory reaction of muscles after electrical stimulation of sensory fibers) amplitudes during stepping on a slackline indicated a more anticipatory/feedforward organization of postural control. Another explanation relates to the central nervous plane.^{30,31} For example, Gebel et al.³⁰ used electroencephalographic analyses to examine the impact of increasing balance task difficulty on cortical activity in adolescents. They observed significant increases in the theta frequency band power and decreases in the alpha-2 frequency band power with increasing levels of task difficulty. The authors interpreted their finding as indicating enhanced attentional processes and/or error monitoring as well as increased sensory information processing due to increasing postural demands. Furthermore, participants in the DBT group exclusively engaged in dynamic balance exercises. As a result, they were primarily assessed for dynamic balance performance, although evaluations of other abilities such as static and quasi-dynamic balance performance were also conducted. Consequently, it would be reasonable to anticipate a 'learning effect' in which improvements in one ability, such as dynamic balance performance, could positively influence other dimensions like static and quasi-dynamic balance performance. This 'learning effect' could explain the greater improvements observed in balance and overall performance following DBT compared to QDBT and SBT among children. The results of the present study demonstrated that DBT yielded in a greater improvement in horizontal and vertical jump performance (i.e., SLJ and CMJ height), in children. Recently, Gebel et al.,³ in their narrative review, showed that BT has the potential to induce

transfer effects to selected health parameters (e.g., muscle strength/power) in youth. The authors argued that BT-induced neuromuscular adaptations might selectively be beneficial for explosive type of muscle actions (i.e., CMJ height and SLJ). Results from the present study could be explained by the fact that the test measures were performed within a more limited base of support (CMJ test) or with the center of gravity being moved outside the base of support (SLJ, CMJ), which would be affected to a much greater extent by DBT. In explaining the present results, it is important to acknowledge that we studied prepubertal children, who frequently performed dynamic balance tasks (i.e., tandem and hopping and landing with bipedal and unipedal) in their daily life activities and their physical exercise lessons. Since precise control of postural sway is needed for dynamic movement, children with enhanced control of postural sway can better control their body position or center of mass over their BoS.^{2,12} Furthermore, the present results may be attributed to the improved proprioceptive afferent feedback and the anticipatory postural adjustments^{2,30} and motor coordination and to the decreased postural sway parameters (greater stability) among other balance-related mechanisms.

Furthermore, as maturation has been shown to contribute to the trainability of balance,^{2,30} the rate at which these individuals are experiencing maturation may, in part, explain the superiority DBT-responses found in our study. Hence, using high balance training task difficulty (i.e., DBT), children with immature balance capabilities and less developed neuromuscular system¹² would conceivably respond positively to DBT. Because children's motor skills are still emerging,^{2,32} this could be related to the general adaptive mechanisms that occur in the level of the spinal and supraspinal centers following DBT in children.

Although we present a novel addition to the literature, this study is not without limitations, and should be mentioned, accordingly. First, the sample of participants involved comprised pre-pubertal school children. Therefore, the results obtained may only be generalized to similar samples of participants. Second, balancing on the stabilometer is a task with high internal, but low external validity, which limits the

transferability of the present results on balance tasks performed during everyday life or sports. Moreover, future studies need to elucidate cortical activity during the performance of balance tasks, with increasing difficulty, using robust measures, such as electroencephalography. Lastly, we were unable to quantify the level of fatigue immediately following the implementation of the balance training (BT) exercise programs, which could potentially have influenced the observed effects on balance and muscle power. Future studies could include assessments of Ratings of Perceived Exertion (RPE) within a few minutes after the execution of Static Balance Training (SBT), Quasi-Dynamic Balance Training (QDBT), and Dynamic Balance Training (DBT) exercises to determine whether fatigue may have played a role in mitigating the observed effects.

6. Conclusion

The present study, to the best of our knowledge, is the first to investigate the effects of Static Balance Training (SBT), Quasi-Dynamic Balance Training (QDBT), and Dynamic Balance Training (DBT) on static, quasi-dynamic, and dynamic balance, as well as lower limb muscle power in prepubertal children.

In summary, the most relevant findings indicate that DBT resulted in significantly greater improvements in balance and muscle power performance compared to QDBT and SBT. These results suggest that prepubertal children exhibit heightened responsiveness to DBT for enhancing balance and power performance. This insight can be instrumental in optimizing individualized balance training regimens and fostering long-term motor skill development in this age group.

As balance and muscle power performance are often integrated into physical education courses, the findings of this study hold potential for incorporation into regular physical education curricula. Scientists and practitioners should consider the outcomes of this research when working with children and implementing balance exercises. Consequently, for those aiming to improve balance and jumping performance in children, Dynamic Balance Training (DBT) should be taken into account.

Although our study involved children in a physical education class, it is conceivable that Static Balance Training (SBT) and Quasi-Dynamic Balance Training (QDBT) exercises might be better suited for obese children due to their lower training loads during balance training. Further research in this area is warranted.

CRediT authorship contribution statement

R.H., L.A. and H.R. were involved in conceptualization and methodology. R.H. and L.A. conducted the study. P.G., J.G.M. and J.C.C. analyzed and interpreted the data. R.H. drafted the manuscript. H.R., J.G.M. and J.C.C. critically revised the manuscript. All authors have read and approved the final version of the manuscript.

Funding information

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Confirmation of ethical compliance

The study was conducted according to the latest version of the Declaration of Helsinki and the protocol was fully approved by the Local Clinical Research Ethics Committee “Personal Protection Committee” under the following code (N°: 0365//2021) before the commencement of the assessments. Written informed consent to participate in this study was provided by the participants’ legal guardian/next of kin.

Declaration of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgments

We would like to thank the participants and the research assistants involved in the data collection.

Appendix 1. CONSORT 2010 checklist of information to include when reporting a randomized trial¹

Section/topic	Item no	Checklist item	Reported on page no
Title and abstract	1a	Identification as a randomized trial in the title	2
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	4–19
Introduction Background and objectives	2a	Scientific background and explanation of rationale	36–83
	2a. o	Explain the mechanism (e.g., pathophysiological, pharmacological, etc.) or theoretical framework by which the intervention is expected to cause change in the outcome	36–75
	2b	Specific objectives or hypotheses	76–83
Methods Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	99–106
	3b	Important changes to methods after trial commencement (such as eligibility criteria), with reasons	N/A
Participants	4a	Eligibility criteria for participants	93–95
	4b	Settings and locations where the data were collected	108–109, 125–126
	4b. o	Explain suitability of setting(s) (e.g., measuring blood pressure in clinic vs. home) where the outcome data was collected, if applicable	125
Interventions	5	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered	125–128, 209–231
Outcomes	6a	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed	151–206
	6a. 1	Provide a rationale for the selection of the domain for the trial’s primary outcome	61–69
	6a. 2	Describe the specific measurement variable (e.g., systolic blood pressure), analysis metric (e.g., change from baseline, final value, time to event), method of aggregation (e.g., mean, proportion), and the time point for each outcome	128–129, 153–154, 156–158, 169–173, 197–198, 200–202, 238–242
	6a. 3	If the analysis metric for the primary outcome represents within-participant change, define and justify the minimal important change in individuals	N/A

(continued on next page)

¹ We strongly recommend reading this statement in conjunction with the CONSORT 2010 Explanation and Elaboration for important clarifications on all the items. If relevant, we also recommend reading CONSORT extensions for cluster randomized trials, non-inferiority and equivalence trials, non-pharmacological treatments, herbal interventions, and pragmatic trials. Additional extensions are forthcoming: for those and for up to date references relevant to this checklist, see www.consort-statement.org.

(continued)

Section/topic	Item no	Checklist item	Reported on page no
Sample size	6a. 4	If the outcome data were continuous, but were analyzed as categorical (method of aggregation), specify the cutoff values used	N/A
	6a. 5	If outcome assessments were performed at several time points after randomization, state the time points used for the analysis	136–137
	6a. 6	If a composite outcome was used, define all individual components of the composite outcome	N/A
	6a. 7	Identify any outcomes that were not prespecified in a trial registry or trial protocol	N/A
	6a. 8	Provide a description of the study instruments used to assess the outcome (e.g., questionnaires, laboratory tests) along with reliability, validity, and responsiveness in a population similar to the study sample	125–206, 260–264
	6a. 9	Describe who assessed the outcome (e.g., nurse, parent) and any qualifications or trial-specific training necessary to administer the study instruments to assess the outcome	131–132
	6a. 10	Describe any processes used to promote outcome data quality during data collection (e.g., duplicate measurements) and after data collection (e.g., range checks of outcome data values), or state where these details can be found	153–154, 172–173, 198–199, 205–206
	6a. o	Specify whether the outcome is routinely collected during usual care or specific to purpose of trial	128–129
	6b	Any changes to trial outcomes after the trial commenced, with reasons	N/A
	7a	How sample size was determined	87–92
Randomization: Sequence generation	7a. 1	Define and justify the target difference between treatment groups (e.g., the minimal important difference)	243–244, 246
	7b	When applicable, explanation of any interim analyses and stopping guidelines	113–115
Allocation concealment mechanism	8a	Method used to generate the random allocation sequence	98–99
	8b	Type of randomization; details of any restriction (such as blocking and block size)	99–100
Implementation	9	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned	98–100
Blinding	10	Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions	98, 99
Statistical methods	11a	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how	131–132
	11b	If relevant, description of the similarity of interventions	209–231
	12a	Statistical methods used to compare groups for primary and secondary outcomes	236–247

(continued)

Section/topic	Item no	Checklist item	Reported on page no
Results	12a. 1	Describe any methods used to account for multiplicity in the analysis or interpretation of the primary and secondary outcomes (e.g., co- primary outcomes, same outcome assessed at multiple time points, or subgroup analyses of one outcome)	238–239
	12a. 2	State and justify any criteria for excluding any outcome data from the analysis and reporting, or report that no outcome data were excluded	237–238
	12a. 3	Describe methods to assess patterns of missingness (e.g., missing not at random), and describe the methods to handle missing outcome items or entire assessments (e.g., multiple imputation)	237–238
	12a. 4	Provide definition of outcome analysis population relating to protocol non-adherence (e.g., as randomized analysis)	251–252
	12a. o	Justify inclusion of covariates/factors used for analyzing the outcome data	N/A
	12b	Methods for additional analyses, such as subgroup analyses and adjusted analyses	238–244
	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analyzed for the primary outcome	Fig. 2 Table 2
	13b	For each group, losses and exclusions after randomization, together with reasons	252–253
	14a	Dates defining the periods of recruitment and follow-up	126
	14b	Why the trial ended or was stopped	125–127
Recruitment	15	A table showing baseline demographic and clinical characteristics for each group	Table 2
Baseline data	16	For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups	Table 2
Numbers analyzed	16a. o	Describe number of missing participant-level observations at each time point by treatment group, with reasons	N/A
Outcomes and estimation	17a	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95 % confidence interval)	268–313, Tables 2–4
	17a. 1	Include results for all prespecified outcome analyses or state where results can be found if not in this report	268–313, Tables 2–4
	17b	For binary outcomes, presentation of both absolute and relative effect sizes is recommended	N/A
Ancillary analyses	18	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory	260–313
	18. 1	If there were any analyses that were not prespecified, explain why they were performed	N/A

(continued)

Section/topic	Item no	Checklist item	Reported on page no
Harms	19	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms)	252–253
Discussion			
Limitations	20	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses	391–403
Generalizability	21	Generalizability (external validity, applicability) of the trial findings	405–421
Interpretation	22	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	318–390
Other information			
Registration	23	Registration number and name of trial registry	115–119
Protocol	24	Where the full trial protocol can be accessed, if available	N/A
Funding	25	Sources of funding and other support (such as supply of drugs), role of funders	N/A

Note: o, optional items.

Appendix 2. The TIDieR (Template for Intervention Description and Replication) Checklist²: information to include when describing an intervention and the location of the information

Item number	Item	Where located ^a	
		Primary paper (page or appendix number)	Other ^b (details)
	BRIEF NAME		
1.	Provide the name or a phrase that describes the intervention.	1–2	_____
	WHY		
2.	Describe any rationale, theory, or goal of the elements essential to the intervention.	37–84	_____
	WHAT		
3.	Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	Table 1	_____
4.	Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	126–207	_____

² We strongly recommend using this checklist in conjunction with the TIDieR guide (see *BMJ* 2014; 348: g1687) which contains an explanation and elaboration for each item. The focus of TIDieR is on reporting details of the intervention elements (and where relevant, comparison elements) of a study. Other elements and methodological features of studies are covered by other reporting statements and checklists and have not been duplicated as part of the TIDieR checklist. When a randomized trial is being reported, the TIDieR checklist should be used in conjunction with the CONSORT statement (see www.consort-statement.org) as an extension of Item 5 of the CONSORT 2010 Statement. When a clinical trial protocol is being reported, the TIDieR checklist should be used in conjunction with the SPIRIT statement as an extension of Item 11 of the SPIRIT 2013 Statement (see www.spirit-statement.org). For alternate study designs, TIDieR can be used in conjunction with the appropriate checklist for that study design (see www.equator-network.org).

(continued)

Item number	Item	Where located ^a	
		Primary paper (page or appendix number)	Other ^b (details)
	WHO PROVIDED		
5.	For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	128–131	_____
	HOW		
6.	Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	131	_____
	WHERE		
7.	Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	126–127	_____
	WHEN and HOW MUCH		
8.	Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	127, 133–135, 210–232	_____
	TAILORING		
9.	If the intervention was planned to be personalized, titrated or adapted, then describe what, why, when, and how.	_____	_____
	MODIFICATIONS		
10. ^c	If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	N/A	_____
	HOW WELL		
11.	Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	133	_____
12. ^c	Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	252	_____

^a Authors - use N/A if an item is not applicable for the intervention being described. Reviewers - use '?' if information about the element is not reported/not sufficiently reported.

^b If the information is not provided in the primary paper, give details of where this information is available. This may include locations such as a published protocol or other published papers (provide citation details) or a website (provide the URL).

^c If completing the TIDieR checklist for a protocol, these items are not relevant to the protocol and cannot be described until the study is complete.

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